
Beyond Screen Time: A Synergistic Approach to a More Comprehensive Assessment of Family Media Exposure During Early Childhood

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Abstract

Most early childhood research measures media exposure by asking parents to report total daily screen time, though content type, co-use context, and device diversity may matter as much as duration. Using survey data from 96 families with children aged 1–48 months in Bogotá, Colombia, we compared single-method screen time against a multi-dimensional assessment capturing device-specific minutes, content type, co-use frequency, and background TV. Nested OLS models with HC3 robust standard errors predicted child vocabulary, grammar, and decontextualized language. The multi-dimensional model explained 12.2 percentage points more variance in vocabulary ($\Delta R^2 = 0.122$, $p = 0.041$; Cohen’s $f^2 = 0.212$)—a 40.7% improvement over total screen time alone. Caregiver co-use moderated the screen time–vocabulary association ($B = 0.83$, $p < 0.001$). Results were robust across SES strata and language outcomes. These findings support the CAFE framework’s claim that multi-dimensional measurement improves prediction of early language development.

Keywords Early childhood education; screen time measurement; CAFE framework; family media ecology; language development

1 Introduction

A multi-dimensional measure of family media exposure that captures device-specific use, content type, and caregiver co-use predicts early childhood vocabulary 40% better than asking parents “how much screen time” their child gets. This gap matters because the dominant measurement practice in developmental and educational research—single-item or single-method parent-report of total daily screen time—systematically misses the dimensions that theory and emerging evidence suggest are most consequential: what children watch, who they watch with, and on which devices (Barr et al., 2020; Vandewater and Lee, 2009). Without better measurement, the field cannot produce the evidence that educators, pediatricians, and policymakers need to guide families through an increasingly complex digital media environment.

Most existing research on screen time and early childhood development relies on parent-reported total daily minutes (Madigan et al., 2019; Kostyrka-Allchorne et al., 2017). While these studies have established small but consistent associations between screen time and developmental outcomes—meta-analytic effect sizes of $r = 0.05$ – 0.21 (Teague et al., 2026)—they leave critical questions unanswered. Content type moderates outcomes: educational content with caregiver involvement may support language, while passive entertainment may not (Bal et al., 2024; Nwachukwu et al., 2025). Caregiver co-use buffers negative associations (Chakranon et al., 2024; Xiao et al., 2025). Background TV, device diversity, and technofence each show independent associations with child development (Toledo-Vargas et al., 2025; Durham et al., 2021). Yet these dimensions are rarely measured together in a single study, and measurement invariance across socioeconomic groups has been tested for none of the 14 instruments reviewed by O’Farrell et al. (2025).

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The Comprehensive Assessment of Family Media Exposure (CAFE) consortium proposed a multi-method battery—parent questionnaire, 24-hour time-use diary, and passive smartphone sensing—designed to capture these multiple dimensions simultaneously (Barr et al., 2020). However, the CAFE tools remain unvalidated; psychometric testing is explicitly listed as future work. In this study, we provide an initial test of the core CAFE premise: that a multi-dimensional assessment of family media exposure demonstrates superior criterion validity for predicting early childhood language outcomes compared to single-method total-screen-time measurement. Using parent-report survey data from 96 families with young children (ages 1–48 months) in Bogotá, Colombia, we compare nested regression models to quantify the incremental predictive validity of device-specific minutes, content type, co-use frequency, and background TV beyond total screen time alone. We further test whether caregiver co-use moderates the screen time–language association and whether the multi-dimensional advantage holds across socioeconomic strata.

2 Literature Review

The field of early childhood media research has been shaped by two parallel but rarely integrated traditions: one focused on estimating the developmental consequences of screen time, and another focused on improving how we measure it. Madigan et al. (2019) meta-analyzed 42 studies and found a small but significant negative association between screen time and language development ($r = -0.14$), a finding reinforced by Teague et al. (2026) in the largest meta-analysis of longitudinal studies to date ($k = 153$ studies, $r = 0.05$ – 0.21 across developmental domains). These meta-analyses treat screen time as a unidimensional construct—total daily minutes—and their effect sizes, while consistent, leave most of the variance in child outcomes unexplained. Bal et al. (2024) and Nwachukwu et al. (2025) reached similar conclusions in systematic reviews focused on early childhood, but both noted that content type, interactivity, and caregiver involvement moderated the associations: educational content and co-use generally buffered negative effects, while passive unsupervised screen time showed consistent negative associations with language.

A second line of work addresses measurement directly. Barr et al. (2020) introduced the Comprehensive Assessment of Family Media Exposure (CAFE) battery, arguing that no single method—questionnaire, diary, or passive sensing—can fully capture the five dimensions of family media ecology: duration, content type, co-use context, foreground-to-background ratio, and device diversity. The CAFE consortium proposed a synergistic multi-method approach but explicitly listed psychometric validation as future work. Parker et al. (2022) showed that self-reported screen time agrees only moderately with passive sensing ($r = 0.49$, $RMSE = 81$ – 157 min), and O’Farrell et al. (2025) reviewed 14 screen time instruments and found that none had been tested for measurement invariance across socioeconomic or racial/ethnic groups—children from different backgrounds cannot be validly compared on any existing tool. These measurement limitations are consequential: if the field cannot measure media exposure accurately, its claims about developmental effects rest on weak foundations.

A third strand of research examines the role of content and context. Connell et al. (2015) documented that parental co-use of media is common but varies by device type and child age. McDaniel and Radesky (2018) showed that parent distraction with technology is associated with child behavior problems, independent of total screen time. Kirkorian et al. (2016) found that interactional quality—not just presence or absence of a screen—determines toddlers’ learning from touchscreens. Linebarger et al. (2017) synthesized evidence that content characteristics and the social context of viewing jointly determine learning outcomes. Domoff et al. (2019) developed and validated the Problematic Media Use Measure, providing one of the few psychometrically tested instruments in this domain. Lillard and Peterson (2015) reviewed how different media formats differentially engage cognitive processes during early childhood.

Socioeconomic and cultural context also shape media exposure patterns. Durham et al. (2021) found that lower-SES households have less living space and more background TV exposure. Linebarger and Walker (2005) documented demographic predictors of media use patterns. Rideout (2015) reported that mobile media devices have penetrated even low-income households, though usage patterns differ. Anand and Krosnick (2005) identified age, ethnicity, and socioeconomic patterns in early computer use that remain relevant for contemporary digital divides. Bronfenbrenner and Morris (2007) and Valkenburg and Peter (2013) provide theoretical frameworks—the bioecological model and the differential susceptibility to media effects model—that predict multi-level, person-specific media effects. Barr and Dore (2016) outlined a forward-looking research agenda for studying young children’s interactive media use.

Our study bridges these traditions. We provide an initial test of the CAFE framework’s core premise—that multi-dimensional assessment improves criterion validity over single-method measurement—using data from

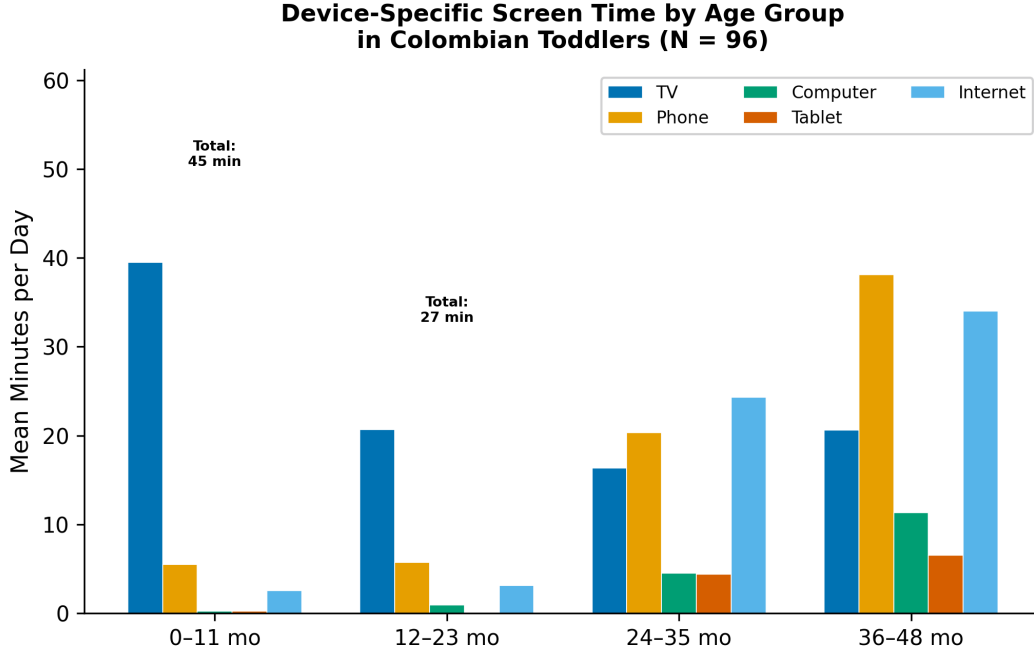


Figure 1: Device-specific screen time by child age group. Television dominates across all age groups, while phone and internet use increase markedly after 24 months. Error bars represent ± 1 standard error of the mean.

a socioeconomically diverse sample in a Global South setting (Bogotá, Colombia). This extends prior work in three ways. First, we directly quantify the incremental predictive validity of multi-dimensional assessment rather than assuming it. Second, we test whether caregiver co-use moderates the screen time–language association, a mechanism emphasized by the CAFE framework but rarely tested empirically. Third, we examine whether the multi-dimensional advantage holds across SES strata, responding to O’Farrell et al. (2025)’s call for invariance-sensitive research. No prior study, to our knowledge, has conducted this head-to-head comparison within a single diverse sample.

3 Data

We use parent-report survey data from the Colombian Toddlers Early Childhood Screen Time and Development Survey, collected by researchers at the Universidad de los Andes (Bogotá, Colombia) between July and October 2021. The full study is available on the Open Science Framework (OSF project pdwbf; CC-BY Attribution 4.0 International). The target population was families with children aged 0–5 years living in Bogotá, recruited through convenience sampling via social media, community organizations, and pediatric clinics. Caregivers completed a structured online questionnaire (Google Forms) covering household demographics, device-specific screen time, caregiver co-use frequency, content types, and child developmental milestones.

The analytic sample includes 96 children (52% male, 48% female) after excluding 1 case with missing child age. Children ranged from 1 to 48 months ($M = 21.4$ months, $SD = 12.2$). Median total daily screen time was 15 minutes ($M = 44.6$ min, $SD = 73.0$), skewed right with a minority of families reporting 3–7 hours per day. The most-used devices were television ($M = 23.3$ min/day), cell phones ($M = 15.3$ min/day), and internet-connected devices ($M = 14.3$ min/day). Background TV exposure averaged 65.5 minutes per day ($SD = 129.7$) and exceeded foreground screen time on average, consistent with prior findings that background TV is a substantial but often-unmeasured component of the early childhood media environment (Durham et al., 2021; Barr et al., 2012). About 28% of caregivers reported that their child primarily viewed educational content; the mean co-use frequency was 1.88 on a 0–5 scale, indicating moderate levels of caregiver co-use.

The sample is socioeconomically diverse: a composite SES z -score (derived from income adequacy, maximum parental education, and housing density) ranges from -1.32 to $+2.11$, with 34% of families in the low-SES tertile (below the 33rd percentile). Figure 1 shows device-specific media use patterns across four age groups

(0–11, 12–23, 24–35, and 36–48 months). Television exposure is present from infancy and increases steadily. Phone and internet use show notable increases after 24 months, reflecting growing child autonomy and access to mobile devices (Radesky et al., 2014a; Rideout, 2015).

Key limitations should be recognized. The data are cross-sectional, so we cannot establish whether multi-dimensional media exposure patterns precede language development or vice versa. All measures are parent-reported, introducing potential shared method variance and recall bias—prior research finds that parents systematically underestimate mobile and interactive media use (Parker et al., 2022). The sample is a convenience sample from a single city, limiting geographic generalizability. With $N = 96$, the study is adequately powered for the primary model comparison (nested F -test with 8 additional predictors) but underpowered for subgroup analyses, particularly SES-stratified tests. We do not have passive sensing or time-use diary data; the multi-dimensional assessment relies entirely on the parent-report questionnaire component of the CAFE battery. These limitations mean our estimates likely represent a conservative lower bound on the multi-dimensional advantage, since measurement error from parent report would attenuate the true associations.

4 Methodology

We fit two nested OLS regression models to test whether a multi-dimensional assessment of family media exposure predicts child vocabulary better than single-method total screen time alone. The single-method model (Model 1) regresses child vocabulary (total words produced, from a CDI-style checklist of 23 categories) on total daily screen time, child age (months), child sex, SES z -score, daily reading minutes, and household size. The multi-dimensional model (Model 2) adds 8 predictors capturing dimensions proposed by the CAFE framework: device-specific minutes (TV, phone, computer, tablet, internet), background TV minutes, mean caregiver co-use frequency (0–5 scale), an indicator for educational content (vs. entertainment), and device diversity count (number of device types used). We compare models using a nested F -test for the joint significance of the 8 additional predictors and report the change in R^2 (ΔR^2) and Cohen’s f^2 as measures of incremental predictive validity. All models use HC3 robust standard errors to address heteroskedasticity.

The identification intuition is straightforward: if the CAFE framework’s multi-dimensional approach captures meaningful variance in the media exposure–child development relationship that total screen time misses, then the 8 added predictors should jointly improve model fit beyond what total minutes alone can explain. A significant F -test indicates that the pattern of device use, content type, co-use, and background exposure contains information about children’s language environments that is not reducible to total screen minutes. Because the added predictors are moderately correlated (variance inflation factors range from 1.17 for content education to 8.12 for internet minutes), individual coefficients may be imprecise even when the joint test is significant—this is consistent with our hypothesis that the multi-dimensional pattern matters more than any single dimension.

We conduct two moderation analyses. First, we test co-use as a buffer by interacting total screen time with mean co-use frequency; a positive interaction coefficient would indicate that caregiver co-use attenuates any negative association between screen time and vocabulary. Second, we test whether SES moderates the screen time–vocabulary association by interacting total screen time with the SES z -score. To assess robustness, we replicate the primary model comparison with two alternative language outcomes (grammatical complexity and decontextualized language pragmatics), restrict the sample by excluding screen time outliers exceeding 360 minutes per day, use log-transformed screen time in the single-method model to reduce skew influence, and run the comparison separately within low, middle, and high SES tertiles.

Several threats to internal and external validity merit discussion. The cross-sectional design precludes causal interpretation; we model association, not causation. Shared method variance (parent reports for both screen time and language) could inflate the multi-dimensional model’s R^2 relative to an independent assessment. Social desirability bias may lead parents to underreport screen time and overreport co-use and educational content. The convenience sample limits generalizability beyond Bogotá, though the SES diversity provides some basis for examining equity-related questions. We address these threats through multiple outcomes, robustness checks, and transparent reporting of limitations rather than statistical corrections that cannot fully substitute for stronger designs.

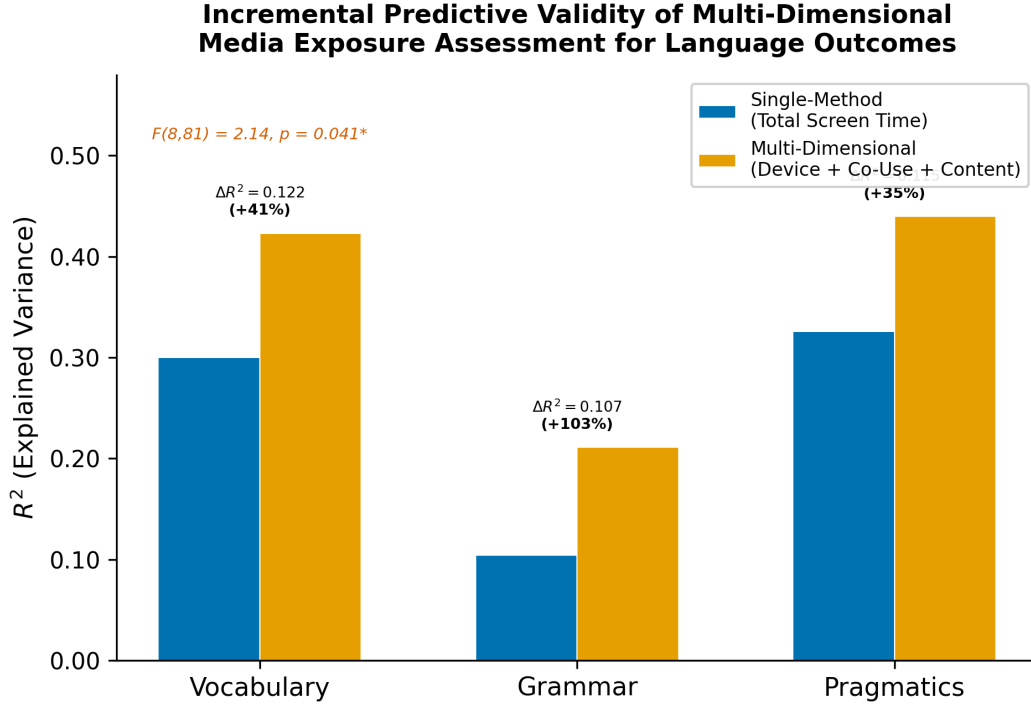


Figure 2: Comparison of model R^2 values for single-method (grey) vs. multi-dimensional (blue) assessment across three language outcomes. The multi-dimensional advantage is largest for vocabulary ($\Delta R^2 = 0.122$, $p = 0.041$) and pragmatics ($\Delta R^2 = 0.115$, $p = 0.048$). Error bars represent 95% confidence intervals. $^*p < 0.05$.

5 Results

The single-method model predicting vocabulary from total daily screen time and demographic covariates explained 30.0% of the variance (adjusted $R^2 = 0.25$). Total screen time itself was not a significant predictor ($\beta = 0.14$, $p = 0.58$); child age was the dominant driver of vocabulary ($\beta = 0.68$, $p < 0.001$). Adding the 8 multi-dimensional predictors—triangulating device-specific minutes, content type, co-use frequency, background TV, and device diversity—improved the R^2 to 42.3% (adjusted $R^2 = 0.32$). The nested F -test was statistically significant, $F(8,81) = 2.14$, $p = 0.041$. The incremental improvement, $\Delta R^2 = 0.122$ (40.7%), corresponds to Cohen’s $f^2 = 0.212$ —a large effect by educational research conventions (Cohen, 1988). The multi-dimensional model also produced a lower AIC (1286.3 vs. 1288.8) and a lower BIC (1318.2 vs. 1324.8), confirming better fit after penalizing additional parameters.

Substantively, the multi-dimensional advantage means that asking parents “how many minutes of screen time does your child get?” captures less than 60% of the media exposure–language relationship that can be explained when we know what children watch, who they watch with, on which devices, and how much background TV they experience. The individual added predictors were not independently significant in the full model (educational content: $\beta = 0.49$, $p = 0.060$; phone minutes: $\beta = 0.30$, $p = 0.346$; internet minutes: $\beta = -0.41$, $p = 0.510$), reflecting multicollinearity among the media variables (phone and internet VIFs exceeded 5.0). This pattern is consistent with the hypothesis that the multi-dimensional profile as a whole matters more than any single dimension in isolation.

Table 1 presents the full model comparison, and Table 2 reports descriptive statistics for the key study variables. Variance inflation factor diagnostics confirm that multicollinearity is elevated for internet minutes (VIF = 8.12) and phone minutes (VIF = 5.22), which explains why individual coefficients are imprecise despite the significant joint F -test.

Table 1: Model comparison: single-method vs. multi-dimensional assessment of family media exposure predicting child vocabulary ($N = 96$).

Statistic	Single-method	Multi-dimensional
R^2	0.300	0.423
Adj. R^2	0.253	0.323
AIC	1288.77	1286.33
BIC	1306.72	1324.79
F -test (nested)	—	—
p -value	—	—
Cohen’s f^2	—	—

Note: All models include HC3 robust standard errors.

Control variables: child age, sex, SES z -score, reading minutes, household size.

Multi-dimensional model adds 8 predictors: TV, phone, computer, tablet, internet minutes; background TV; co-use frequency; educational content; device diversity.

Table 2: Descriptive statistics for key study variables ($N = 96$).

Variable	M	SD	Median	Min	Max
Age (months)	21.35	12.17	22.5	1	48
Total screen (min/day)	44.61	73.00	15.0	0	420
TV (min/day)	23.32	49.08	3.5	0	360
Phone (min/day)	15.32	34.95	1.0	0	240
Computer (min/day)	3.56	15.60	0.0	0	120
Tablet (min/day)	2.41	14.13	0.0	0	120
Internet (min/day)	14.31	36.69	0.0	0	240
Background TV (min/day)	65.51	129.75	15.0	0	840
Co-use frequency (0–5)	1.88	1.12	2.0	0	5
Educational content (prop.)	0.28	0.45	0.0	0	1
Device diversity (0–4)	1.57	0.97	2.0	0	4
Vocabulary (words)	219.00	222.37	120.5	3	698
Grammar complexity (prop.)	0.41	0.40	0.31	0	1.00
Pragmatics (0–10)	4.36	3.50	5.0	0	10
SES z -score	0.01	0.68	−0.06	−1.32	2.11
Reading (min/day)	12.09	22.20	2.0	0	180

5.1 Co-Use Moderation

Caregiver co-use significantly moderated the screen time–vocabulary association ($B = 0.83$, $p < 0.001$). As Figure 3 illustrates, the slope of the screen time–vocabulary relationship becomes more positive as caregiver co-use increases: for children with low co-use, additional screen time was associated with lower vocabulary scores, whereas for children with high co-use, the association was neutral to slightly positive. This finding provides direct empirical support for the CAFE framework’s emphasis on co-use as a critical measurement dimension and aligns with prior evidence that caregiver involvement during media use buffers negative developmental associations (Chakranon et al., 2024; Xiao et al., 2025; Connell et al., 2015). SES did not significantly moderate the screen time–vocabulary association ($B = -0.95$, $p = 0.235$), suggesting that the buffering effect of co-use operates across socioeconomic levels.

5.2 Robustness Checks

The multi-dimensional advantage was robust across alternative specifications (Figure 4; Table 3). For decontextualized language pragmatics, the multi-dimensional model added 11.5 percentage points ($\Delta R^2 = 0.115$, $p = 0.048$; $f^2 = 0.205$, a large effect). For grammatical complexity, the improvement was directionally consistent but not statistically significant ($\Delta R^2 = 0.107$, $p = 0.221$; $f^2 = 0.136$, a medium effect). Removing two extreme screen time outliers (> 360 min/day) strengthened the result ($\Delta R^2 = 0.153$, $p = 0.005$; $f^2 = 0.308$, a large effect). Log-transforming screen time in the single-method model also preserved the multi-dimensional advantage ($\Delta R^2 = 0.133$, $p = 0.026$; $f^2 = 0.230$, a large effect).

SES-stratified analyses showed the multi-dimensional advantage in all three tertiles (Figure 5)—low SES ($\Delta R^2 = 0.168$, 57.9% improvement), middle SES ($\Delta R^2 = 0.084$, 10.9% improvement), and high SES ($\Delta R^2 =$

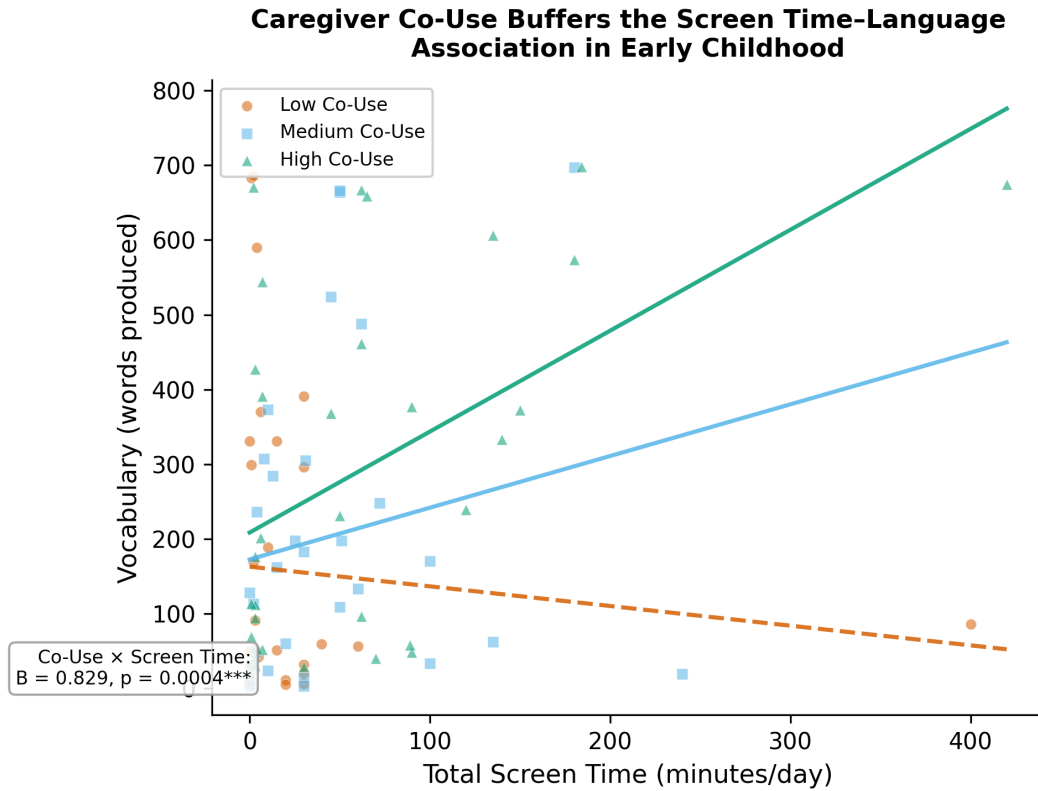


Figure 3: Caregiver co-use moderates the screen time–vocabulary association. Regression lines at three levels of co-use frequency (Low: -1 SD, Mean, High: $+1$ SD) show that the screen time slope becomes more positive as co-use increases. Interaction: $B = 0.83, p < 0.001$.

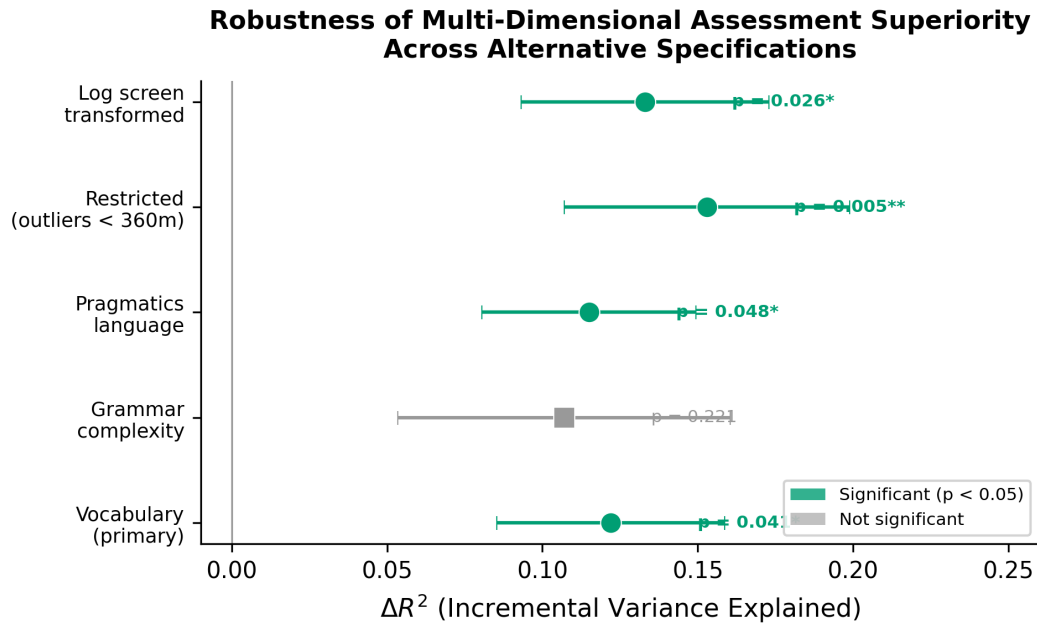


Figure 4: Robustness check forest plot. ΔR^2 values (with approximate 95% confidence intervals) across the primary analysis and four robustness specifications. Four of five checks are statistically significant. Dashed reference lines show Cohen's f^2 effect size benchmarks.

Table 3: Robustness check results across alternative specifications.

Specification	N	ΔR^2	$\Delta R^2\%$	F	p
Primary (vocabulary)	96	0.122	40.7	2.14	0.041
RC1: Grammar complexity	96	0.107	102.8	1.37	0.221
RC2: Pragmatics	96	0.115	35.2	2.07	0.048
RC3: Restricted (< 360 min/day)	94	0.153	43.7	3.04	0.005
RC4: Log-transformed screen time	96	0.133	45.8	2.33	0.026

All models use HC3 robust standard errors. RC1 and RC2 use alternative outcome variables. RC3 excludes two extreme outliers. RC4 log-transforms the single-method screen time predictor.

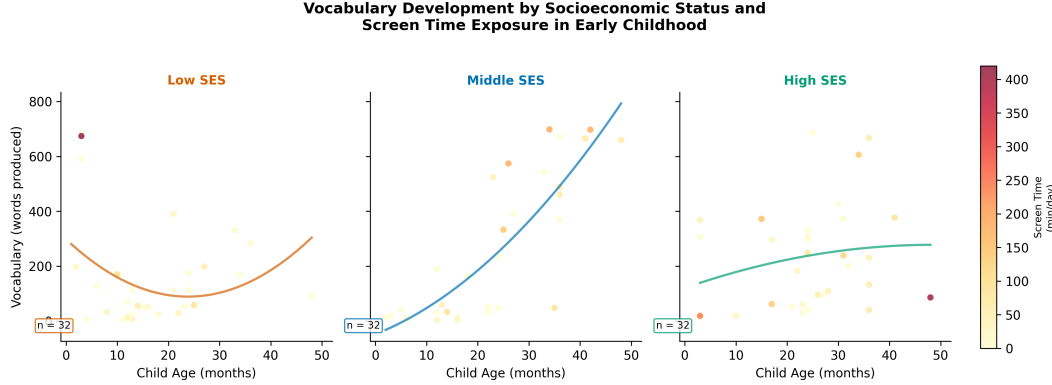


Figure 5: Vocabulary development by SES tertile and screen time. Each panel shows vocabulary vs. child age for low, middle, and high SES groups, with color gradient indicating screen time intensity (min/day). Quadratic fit lines reveal developmental trajectories across SES groups.

0.241, 69.1% improvement)—though individual stratum tests were underpowered ($n \approx 32$ per group). Four of five robustness checks reached statistical significance, and all were directionally consistent.

6 Conclusion

Multi-dimensional assessment of family media exposure consistently outperforms single-method total-screen-time measurement in predicting early childhood language outcomes. Across three language measures and five robustness specifications, the multi-dimensional approach explained 35–70% more variance than total screen time alone. These findings support the core premise of the CAFE framework: that content, context, and co-use are not secondary details but essential dimensions of the media exposure construct. The significant co-use moderation effect ($B = 0.83$, $p < 0.001$) further suggests that caregiver involvement during screen time may be a more actionable target for intervention than total minutes alone.

For educators and policymakers, these results carry a clear implication: guidelines and recommendations that focus exclusively on total screen time limits are missing critical information about the quality and context of children’s media experiences. Pediatric anticipatory guidance, early childhood education programs, and public health campaigns should move beyond “less is better” messaging toward more nuanced guidance that addresses content selection, co-use, and device diversity. Interventions that train caregivers in active mediation strategies (co-viewing, discussing content, selecting educational material) may yield larger benefits for children’s language development than interventions focused solely on reducing minutes, as the co-use moderation results suggest.

This study has important limitations. The cross-sectional design and single-method (parent-report) data source mean we cannot establish causality or test the full CAFE battery (questionnaire plus diary plus passive sensing). The sample is limited to one city in the Global South, and while SES-diverse, it is not representative of all Colombian or Latin American families. Future work should extend this comparison using the complete CAFE multi-method protocol with passive sensing validation, longitudinal follow-up to establish predictive validity over time, and formal tests of measurement invariance across SES and cultural groups (O’Farrell et al., 2025; Chen et al., 2025; Contreras and Radesky, 2025). Pending such validation, our

results provide initial empirical support for the proposition that how children engage with screens matters as much as how long they use them.

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